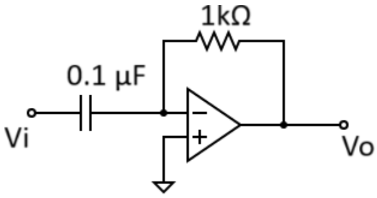
INDIAN INSTITUTE OF TECHNOLOGY DELHI
INTRODUCTION TO ELECTRICAL ENGINEERING
| AELL101 QUIZ-3 |

NAME: **ENTRY NO:**
DATE: 23-04-2025 (9 AM – 9:45 AM) **DURATION: 45 min**
Total marks: 15

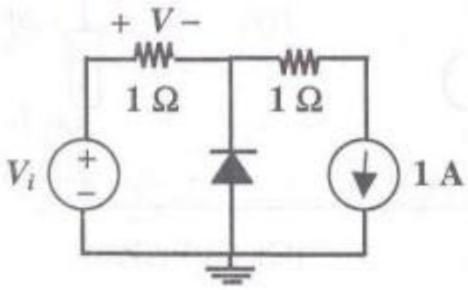
- ✓ **Each question carries equal marks. Space for rough work is given on the last page.**
- ✓ **MCQ (single choice): CIRCLE correct option, otherwise no marks.**

1. The operational amplifier shown in the figure is ideal. The input voltage (in Volt) is $V_i = 2\sin(2\pi\times2000t)$. The output voltage Vo is:



A. $-2\cos(2\pi\times2000t)$ B. $8000\,\pi\cos(2\pi\times2000t)$
C. $2\cos(2\pi\times2000t)$ D. $-0.8\,\pi\cos(2\pi\times2000t)$

2. In the circuit below, the diode is ideal. The voltage V is given by

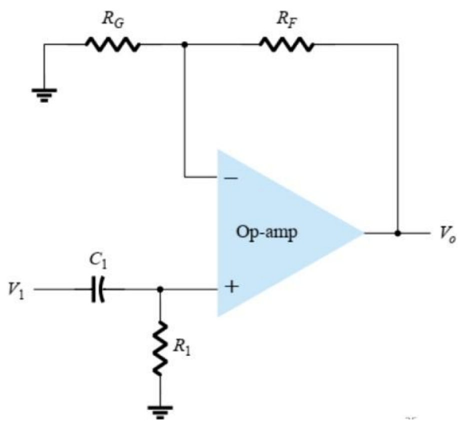


A. $\min(V_i, 1)$ B. $\max(V_i, 1)$ C. $\min(-V_i, 1)$ D. $\max(-V_i, 1)$

3. Which of the following statements are TRUE about OPAMP?
P: Input resistance of an ideal OpAmp is zero
Q: Input resistance of an ideal OpAmp is infinity
R: OpAmp operation in positive feedback configuration is typically unstable
S: OpAmp operation in positive feedback configuration is typically stable

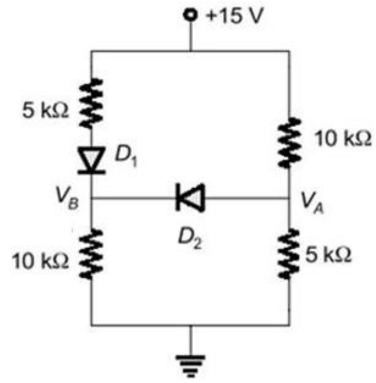
A. P and R B. P and S
C. Q and R D. Q and S

4. What operation does the following OpAmp circuit performs ?



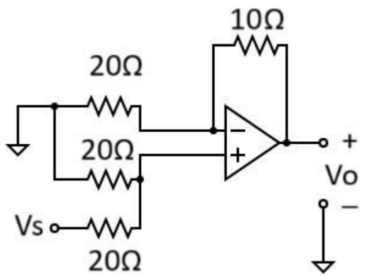
A. Low pass filter B. High pass filter
C. Summer D. None of these

5. Which of the following is correct for the circuit shown on the right, assuming the voltage drop across the forward-biased diodes is 0.7 V?



A. $V_A = 7.62$ V
B. $V_A = -5$ V
C. $V_B = 9.53$V
D. $V_B = 6.92$ V

6. In the given circuit, the output voltage, Vo in terms of Vs is given by

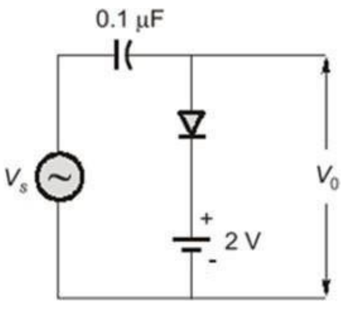


A. 0.75 Vs
B. 1.5 Vs
C. 1.33 Vs
D. 0.66 Vs

7. Which of the following statements are TRUE?
P: A regular (rectifier) diode acts as an open switch in reverse bias condition
Q: A Zener diode can conduct in reverse bias condition.

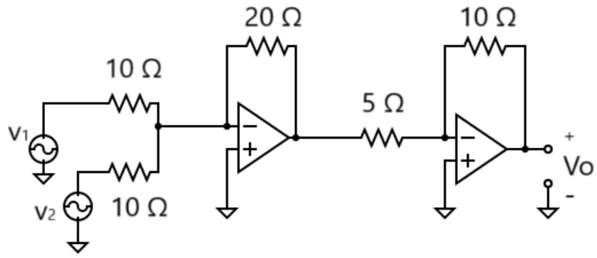
A. Only Q
B. Both P and Q
C. Only P
D. None of these

8. For an input of Vs = 5 sinωt (assuming the ideal diode), the circuit shown below will behave as a:



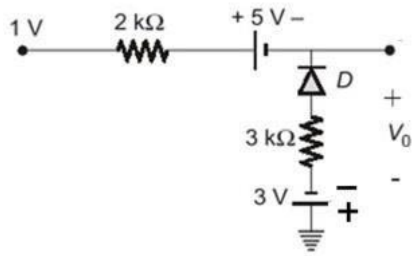
A. Clamper, the signal is shifted up by 3V
B. Clamper, the signal is shifted down by 3V
C. Clamper, the signal is shifted down by 5V
D. Clamper, the signal is shifted up by 5V

9. For the circuit shown below with ideal operational amplifiers, the input voltages are $V_1 = 2V$ and $V_2 = 3V$. The output voltage Vo is:



Ans: _______________

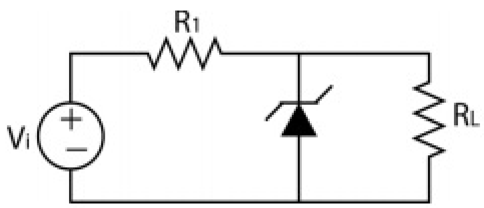
10. Assuming an ideal diode, what is the output voltage Vo for the circuit below?



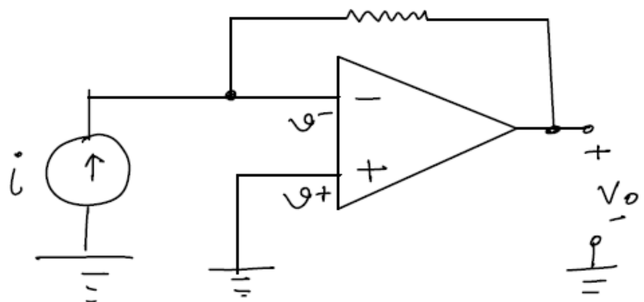
Ans: _______________

11. In the circuit shown below, the breakdown voltage and the maximum current of the Zener diode are 20 V and 60 mA, respectively. The values of R_1 and R_L are 200Ω and 1kΩ respectively. What is the maximum value of Vi that will maintain the Zener diode in the 'on' state?

Ans: __________________

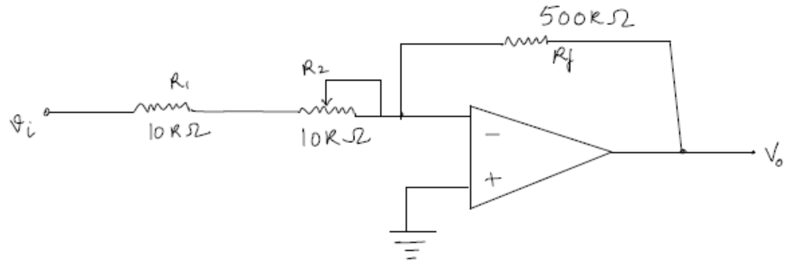


12. For the given circuit, determine Vo in volts, given $i = 2\,A$ and $R = 4\,k\Omega$



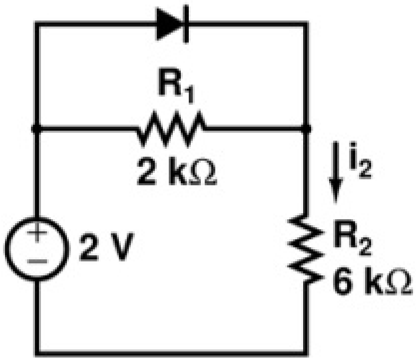
Ans: _______________________

13. What will be the voltage gain range (max and min) for the circuit shown below, given R2 can be adjusted between 0 and 10 $k\Omega$



Ans: min:___________ max: ___________

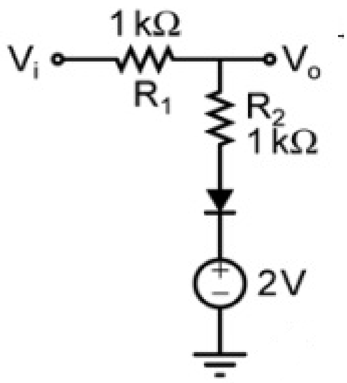
14. Assume that the diode in the figure has V_T= 0.7 V. The magnitude of the current i_2 (in mA) is equal to _____



Ans: ________________________

15. The diode in the circuit shown below has V_T = 0.7 Volts. If Vi=5sinωt Volts, the minimum and maximum values of Vo (in Volts) are, respectively _____ and _______,

Ans: _____________ __________________



Rough work space: